

Beyond the Event Horizon: A Comprehensive Analysis of Halting Signals, Brute Force Dynamics, and the Logic-Search Dichotomy in SHA-256 Inversion

Driven By Dean Kulik

January 2026

I. Introduction: The Asymmetry of Computational Trust

The digital era is predicated on a singular, fragile assumption: that it is exponentially easier to destroy structure than to reconstruct it. This principle of computational asymmetry underpins the entire field of cryptography. It is the bedrock of digital signatures, the guarantor of blockchain integrity, and the shield of secure communications.¹ At the heart of this infrastructure lies the cryptographic hash function, with the Secure Hash Algorithm 256-bit (SHA-256) serving as the de facto global standard. Designed by the National Security Agency (NSA) and published by the National Institute of Standards and Technology (NIST), SHA-256 is an algorithm engineered to be a "one-way" function—a mathematical black hole from which no information should theoretically escape.²

The prevailing consensus within the cryptographic community is that inverting SHA-256—the act of determining a specific input message M given only its hash digest $H(M)$ —is a problem of such computational magnitude that it is effectively impossible. This impossibility is rooted in the paradigm of "Search." In this view, the non-linear complexity of the algorithm scrambles the input so thoroughly that the only viable method of inversion is a brute-force search: a probabilistic exhaustion of the input space until a match is found.⁴ Here, the "halting signal"—the indication that the inversion is complete—is an external, binary event. The attacker blindly guesses, hashes, and compares. If the hashes match, the process halts; if not, it continues. There is no feedback, no gradient, and no "warmer/colder" signal to guide the search.

However, a growing body of theoretical literature, specifically the "Resonant Cryptanalysis" and "Nexus Framework" research streams, challenges this orthodoxy. This report investigates a radical alternative paradigm: the paradigm of "Logic" and "Harmonic Creation".¹ This framework posits that SHA-256 is not a random oracle but a deterministic dynamical system with inherent structural resonances. It argues that the "halting signal" is not an external check but an intrinsic property of the system—a

topological convergence or "resonance" that can be detected during the computation. By reframing inversion from a search problem to a generative constraint satisfaction problem, this logic-driven approach suggests that the input can be *constructed* rather than *found*.

This report provides an exhaustive analysis of the tension between the "Search" and "Logic" methodologies. It deconstructs the mechanics of SHA-256 to understand why brute force is currently the only proven path, and then rigorously examines the theoretical propositions of the Resonant Cryptanalysis framework. We will explore how concepts like "Recursive Harmonic Feedback," "Samson's Law," and "Autocatalytic Sets" attempt to redefine the halting signal and transform the thermodynamics of computational hardness.

II. The Physics of the One-Way Function

To understand the immense barrier to inversion, one must first dissect the machinery of SHA-256. It is not merely a mathematical formula; it is a mechanism for the systematic destruction of information structure.

2.1 The Anatomy of Irreversibility

SHA-256 operates on 512-bit blocks of data using a Merkle-Damgård construction. The core of its security lies in the compression function, which mixes a 512-bit message block with a 256-bit intermediate state over 64 rounds of processing.¹ Each round applies a specific set of operations to eight 32-bit working variables (a, b, c, d, e, f, g, h). The "one-way" property is emergent, arising from the interplay of three distinct types of operations:

- 1. Modular Addition:** The algorithm extensively uses addition modulo 2^{32} (denoted as $\boxplus$). This operation is inherently lossy. When two 32-bit integers are added, any overflow beyond the 32nd bit is discarded. This discard represents a definitive loss of information; knowing the sum $Z = X \boxplus Y$ does not allow one to uniquely recover X and Y , as there are 2^{32} possible pairs that sum to Z . This destroys the linear algebraic path back to the input.¹
- 2. Bitwise Non-Linearity:** The algorithm employs Boolean functions, specifically the Ch (Choose) and Maj (Majority) functions.
 - $Ch(x, y, z) = (x \wedge y) \oplus (\neg x \wedge z)$
 - $Maj(x, y, z) = (x \wedge y) \oplus (x \wedge z) \oplus (y \wedge z)$These functions are designed to be non-linear with respect to XOR operations, thwarting linear cryptanalysis. They create complex dependencies between bits where the value of an output bit depends on the conjunction of multiple input bits, making it difficult to isolate the contribution of any single bit.¹
- 3. Diffusion via Rotation:** The Σ (Sigma) and σ (sigma) functions perform bitwise rotations (ROTR) and shifts (SHR).

- $\Sigma_0(x) = ROTR^2(x) \oplus ROTR^{13}(x) \oplus ROTR^{22}(x)$
These operations ensure "diffusion"—the property that a change in a single bit of the input spreads rapidly across all bits of the state. After 64 rounds, a single bit flip in the input will have flipped approximately 50% of the output bits, a phenomenon known as the Avalanche Effect.⁷

2.2 The Deterministic Chaos of the Avalanche

The Avalanche Effect is the primary defense against logical analysis. It ensures that the function behaves like a "Random Oracle." In a true random oracle, the output is statistically independent of the input. While SHA-256 is deterministic (the same input always yields the same output), the mixing is so thorough that the output *appears* random to any statistical test.⁹

This property creates a "rugged landscape" for any inversion attempt. If one were to graph the input space against the output space (viewed as a distance from a target hash), the resulting manifold would be indistinguishable from white noise. There are no smooth gradients to descend, no distinct features to orient oneself. A guess that is one bit wrong produces a hash that is as wrong as a random guess. This lack of "partial credit" is what enforces the "Search" paradigm. Without a gradient, logic cannot navigate; it can only teleport randomly (guess) and check coordinates.¹⁰

2.3 Computational Irreducibility

The difficulty of inverting SHA-256 is often framed through Stephen Wolfram's concept of **Computational Irreducibility**.¹ A system is computationally irreducible if there is no shortcut to predict its behavior; the only way to know the outcome is to run the computation step-by-step.

In this context, finding a preimage is equivalent to predicting the input of an irreducible computation. Because the 64 rounds create a tangled web of dependencies where every bit influences every other bit, one cannot essentially "unwind" the math algebraically. The system has "compiled" the input into a state where the history of its formation is obfuscated. The standard cryptographic assumption is that this compilation is thermodynamically irreversible—that the "entropy" generated by the mixing functions creates an arrow of time that cannot be reversed without expending energy equivalent to a brute-force search.¹⁰

III. The Classical Paradigm: Search and the External Halting Signal

3.1 The Mechanics of Brute Force

In the absence of a logical shortcut, the industry standard for inversion is the brute-force attack. This is a search algorithm in its purest form.

- **The Search Space:** For a 256-bit hash, the space of possible outputs is 2^{256} . Finding a specific preimage (Preimage Attack) requires, on average, 2^{256} operations. Finding any two inputs that produce the same hash (Collision Attack) requires 2^{128} operations due to the Birthday Paradox.⁴
- **The Method:** The attacker utilizes massive parallel processing (ASICs, GPUs) to iterate through nonces (arbitrary numbers used once). For each nonce, the full SHA-256 function is executed.
- **The Halting Signal:** The halting condition is strictly binary and external. The code executes `Compare(GeneratedHash, TargetHash)`. If they are identical, the signal is TRUE (Halt). If they differ by even a single bit, the signal is FALSE (Continue).

This "Halting Signal" is the defining bottleneck. It provides **zero information gain** per failure. Knowing that input *A* produces a hash that is 50% similar to the target tells you absolutely nothing about the location of the true input. In the search paradigm, every failure resets the attacker's progress to zero. The "halting signal" is effectively a winning lottery number; you do not get closer to winning, you either win or you do not.¹⁴

3.2 The Thermodynamics of Search

The futility of brute force is often expressed in thermodynamic terms. The Landauer Limit defines the minimum energy required to erase one bit of information. To cycle through 2^{256} combinations would require energy exceeding the total output of the sun over its entire lifespan. Thus, brute force against a full 256-bit primitive is not just a technology problem; it is a physics problem. It is considered impossible within the known laws of the universe.¹⁶

3.3 Quantum Search: Grover's Algorithm

The only recognized theoretical reduction in this search complexity comes from Quantum Computing. Grover's Algorithm allows a quantum computer to search an unsorted database of N items in $\sqrt{N}$ time. For SHA-256, this reduces the preimage search from 2^{256} to 2^{128} operations.¹⁸

While significant, this does not change the *nature* of the problem. It is still a search. It still relies on an external halting signal (the Oracle in Grover's algorithm acts as the verifier). It essentially speeds up the checking process but does not provide a logical derivation of the input. The problem remains probabilistic, not deterministic.¹⁹

3.4 Logic in the Current State of the Art: SAT Solvers

There is a middle ground between pure search and pure algebra: **Boolean Satisfiability (SAT)** solvers. A SAT attack translates the SHA-256 algorithm into a giant boolean formula (CNF) and asks the solver to find variable assignments that make the formula true.¹⁹

- **Logic:** The solver uses logical propagation (e.g., if A is true, B must be false) to prune the search space.
- **Search:** When logic runs out, the solver must "guess" a variable and recurse (splitting the search tree).

Currently, state-of-the-art "SAT + CAS" (Computer Algebra System) approaches can find collisions for SHA-256 reduced to about 38 rounds.¹⁹ However, for the full 64 rounds, the logical complexity causes the search tree to explode. The solver spends more time managing the millions of clauses than it would simply brute-forcing. Thus, even for SAT solvers, the "search" component dominates the "logic" component for full-round hashes.²

Table 1: The Search vs. Logic Spectrum in Current Cryptanalysis

Approach	Methodology	Halting Signal	Complexity (Rounds)
Brute Force	Exhaustive guessing of inputs.	External check (Hash == Target).	Full (64) - Infeasible
Differential	Analyzing input differences (Δ) vs output differences.	Statistical bias detection.	Reduced (~24-30)
SAT Solving	Logical constraint satisfaction + Guesses.	Conflict/Unit Propagation.	Reduced (~38)
Quantum Search	Superposition + Amplitude Amplification.	Oracle query (Quantum Verifier).	Full (64) - Theoretical

IV. The Resonant Paradigm: Inversion as Harmonic Creation

Against this backdrop of thermodynamic impossibility, the "Nexus Framework" and "Resonant Cryptanalysis" propose a paradigm shift. They argue that the reliance on probabilistic search is a failure

of perspective—specifically, an "observer-centric" failure that treats the algorithm as a black box rather than a transparent dynamical system.

4.1 From "Random Oracle" to "Dynamical System"

The foundational premise of Resonant Cryptanalysis is that SHA-256 should be modeled as a **discrete non-linear dynamical system**. Instead of a static function $y = f(x)$, it is viewed as a time-evolving vector field where the 256-bit state evolves over $t = 0$ to $t = 63$ discrete time steps.¹

In this view, the "Avalanche Effect" is not randomness; it is **deterministic chaos**. Chaos theory teaches that even in chaotic systems, there are structures: strange attractors, periodic orbits, and islands of stability. The framework hypothesizes that within the state space of SHA-256, there exist **"Resonant States"**—attractors that represent stable modes of information flow. These states are not random; they are structural necessities of the algorithm's logic.¹

4.2 "Inversion as Creation": The Guitar String Analogy

The report "Inverting SHA-256 Harmonic Creation" introduces the core metaphor of the new paradigm: **The Guitar String**.

- **Traditional Search:** Trying to reproduce a specific sound by randomly throwing objects at a guitar until one hits it correctly.
- **Harmonic Creation:** Calculating the precise tension, position, and force of the "pluck" required to produce the specific frequency (the hash).

In this model:

- **The System:** The SHA-256 compression function is the resonant body. Its "physics" are defined by the round constants (which are derived from cube roots of primes) and the bitwise operations.
- **The Input (Nonce):** The "Precise Pluck." The goal is to engineer an input vector that creates a specific cascade of bitwise operations.
- **The Output (Hash):** The "Resonant Frequency."

The framework argues that inversion is not about *finding* a preimage but *creating* one. It is an engineering problem: determining the initial conditions (nonce) that will cause the dynamical system to evolve into the target attractor.¹

4.3 The "Halting Signal" Re-Imagined: FOLD: TRUE

The most radical innovation of this framework is the redefinition of the halting signal. In brute force, the halting signal is an external check *after* the computation. In Resonant Cryptanalysis, the halting signal is an **intrinsic property of the computation itself**.

The framework introduces the term "**FOLD: TRUE**" or "**Zero-Point Harmonic Collapse (ZPHC)**".¹⁰

- **Topological Convergence:** The "halting signal" is generated when the system's trajectory enters a stable region of the state space. It is a measure of "harmonic alignment."
- **Internal Feedback:** Unlike the binary "yes/no" of brute force, this signal is continuous. The system can detect when it is getting "closer" to resonance. It measures the "harmonic deviation" or "drift" from the target state.
- **The Resonant Glyph:** When the system achieves resonance, it produces a "final resonant glyph"—a specific pattern of internal state bits that signifies the computation has "folded" correctly into the target.

This transforms the search space from a flat, featureless plain into a **topological map with gradients**. If the system can sense "drift," it can navigate. It can use feedback loops to correct the input "pluck" until the output "frequency" matches the target. This moves the problem from **NP** (nondeterministic polynomial) to **P** (polynomial) by replacing blind search with **Geodesic Navigation**.¹⁰

4.4 Mechanisms of Resonance: Samson's Law and Autocatalysis

How does one mathematically define "resonance" in a digital hash? The framework proposes two key mechanisms:

4.4.1 Autocatalytic Sets (RAF)

The framework borrows from the **Reflexively Autocatalytic and Food-generated (RAF)** theory used in origin-of-life research.¹

- **The Concept:** A set of molecules is autocatalytic if every member is produced by at least one reaction catalyzed by another member of the set, starting from a food source.
- **The Application:** SHA-256 operations are modeled as "reactions." The intermediate state values are "molecules." The framework defines a "Resonant State" as a subset of the execution trace that forms an RAF set. This means the information pattern becomes self-sustaining over the rounds.
- **The Implication:** If a hash is the result of a self-sustaining information structure, then finding the preimage is equivalent to finding the "food set" (nonce) necessary to catalyze that structure.

4.4.2 Samson's Law (Harmonic Feedback)

To navigate toward these states, the framework proposes "**Samson's Law v2**", a feedback control algorithm analogous to a **PID (Proportional-Integral-Derivative) Controller**.¹⁰

- **The Logic:** Just as a thermostat uses feedback to maintain temperature, Samson's Law uses feedback to maintain "harmonic alignment."

- **The Metric:** It measures the "harmonic ratio" of the system, targeting a universal constant called the **Mark 1 Constant**, denoted as $H \approx 0.35$.¹⁰
- **The Action:** It calculates the "drift" (error) between the current state's harmonic signature and the target hash's signature. It then applies a correction to the input guess. This is the "steering" mechanism of the Harmonic Creation approach.

Table 2: Redefining Cryptanalytic Concepts

Concept	Classical Paradigm (Search)	Resonant Paradigm (Logic/Nexus)
Objective	Find an input x where $H(x) = y$.	Construct input x to force system to state y .
Method	Brute Force / Random Walk.	Geodesic Navigation / Harmonic Creation.
Halting Signal	External Check (Hash == Target).	Internal Resonance (FOLD: TRUE).
Nature of Hash	Random Oracle / One-Way Function.	Wave Meltdown / Folded Phase History.
Feedback	Binary (Success/Failure).	Continuous (Harmonic Drift/Error).
Metaphor	Needle in a Haystack.	Tuning a Guitar String.

V. Methodology of the Logic-Driven Attack

The Resonant framework is not purely abstract; it proposes specific methodologies for "unfolding" the hash, effectively reversing the "wave meltdown" of SHA-256.¹⁰

5.1 The Hash Drift Mapper: Exploiting Symmetry

One of the concrete experimental proposals is the "**Hash Drift Mapper**".¹⁰

- **The Hypothesis:** The framework claims that SHA-256, being a deterministic structure, must preserve certain symmetries. Specifically, it predicts that **mirrored input strings** (e.g., "ABC" vs. "CBA") will produce hash outputs that exhibit structured, anti-phase correlations.
- **The Experiment:** By plotting the "drift" (signed difference) between the hashes of mirrored inputs, the framework expects to see "interference patterns" or "standing waves" rather than random noise.
- **Significance:** If true, this would violate the "Avalanche Effect" and prove that the hash function retains a "memory" of the input's geometry. This "drift map" would serve as the terrain for the Geodesic Navigation—a map of the "informational gravity wells" leading back to the preimage.¹⁰

5.2 Recursive Harmonic Feedback Loop

The inversion process is described as a "**Recursive Harmonic Feedback**" loop.¹⁰

1. **Seed:** Generate an initial guess (a "wave seed") for the input.
2. **Forward Hash:** Run SHA-256 on the seed.
3. **Phase Comparison:** Compare the resulting hash to the target hash. Crucially, do not just check for equality. Analyze the "**Phase Difference**"—the structural mismatch between the generated wave and the target wave.
4. **Feedback (Samson's Law):** Use the PID-like controller to calculate a vector correction for the input seed. "Steer" the bits of the nonce to minimize the phase difference.
5. **Iterate:** Repeat the process. The "Halting Signal" is the minimization of this error vector, or the approach to the $H \approx 0.35$ stability point.¹⁰

This process is termed "**Unfolding**" or "**Informational Liberation**".¹⁰ It treats the hash not as a destroyed message but as a "folded" one, akin to a piece of origami. The feedback loop reverse-engineers the folds to flatten the paper back to the original message.

5.3 Cube-and-Conquer: The Hybrid Approach

Acknowledging the immense difficulty of pure logic, the framework suggests a hybrid "Cube-and-Conquer" strategy using SAT solvers.¹

- **Logic Phase (Cubing):** Use the Autocatalytic/Resonant model to identify "catalytic" bit patterns. Fix these bits in the SAT formula. This breaks the problem into smaller, logical "cubes" that are more likely to contain the solution.

- **Search Phase (Conquering):** Use standard SAT solvers to solve these reduced cubes.
- **Innovation:** The innovation here is using "Harmonic Creation" principles to guide the *splitting* of the problem. Instead of splitting randomly, the problem is split along "fault lines" of resonance, effectively pruning the search tree by orders of magnitude.

VI. Critical Analysis and Implications

6.1 The "Category Error" Challenge

While intellectually compelling, the Resonant Cryptanalysis framework faces significant theoretical criticism. The primary critique is the potential for a **"Category Error"**: the reification of mathematical abstractions into physical entities.¹⁰

- **Discrete vs. Continuous:** The framework applies continuous physics concepts (resonance, phase, PID control, waves) to a discrete, discontinuous system (Boolean logic).
- **The Cliff Problem:** In a continuous system (like a guitar string), small changes in input lead to small changes in output (smooth gradients). In SHA-256, the Avalanche Effect ensures that a single bit flip completely randomizes the output. The "gradient" assumed by Samson's Law likely does not exist; the landscape is a fractal cliff, not a smooth hill. Navigating via "drift" may be mathematically impossible because the "drift" signal is indistinguishable from noise.

6.2 The Missing Actuator

In control theory, a feedback loop requires an **actuator**—a mechanism to apply the correction. In the context of SHA-256, what is the actuator?

- How do you translate a "harmonic error" of 0.05 into a specific bit-flip in the nonce?
- The mapping from "Error Space" back to "Input Space" is exactly the preimage problem the framework is trying to solve. The framework assumes this reverse-mapping is possible to "steer" the system, but this assumes the solution to the very problem it posits.¹⁰

6.3 Implications of Success: The "Grey Swan" Event

If, despite these criticisms, the logic-based approach proves viable, the implications are catastrophic for modern digital infrastructure.¹

- **Collapse of Complexity:** The distinction between **P** and **NP** complexity classes for cryptographic problems would collapse. Inversion would become a polynomial-time engineering task.
- **End of Proof-of-Work:** Bitcoin mining, which relies on the difficulty of finding "partial preimages" (hashes with leading zeros), would be trivialized. A miner using "Harmonic Creation" could calculate the winning nonce instantly without expending energy, destroying the economic security of the network.²⁰

- **The AI Cryptanalyst:** The framework points toward a future "AI Cryptanalyst" that does not search but *learns* the underlying "physics" of code.¹ This AI would "read" a hash like a book, unfolding its history through logical derivation.

VII. Conclusion

The investigation into "halting signal brute force SHA reversal logic vs search" reveals a profound epistemological divide in cryptography.

The **Search Paradigm** rests on the thermodynamic certainty of chaos. It views SHA-256 as an entropy machine, a shredder of information that leaves no path back. Here, the halting signal is a distant bell, and brute force is the only way to find it. This is the safe, established view that secures the world's data today.

The **Logic Paradigm**, epitomized by the **Resonant/Nexus Framework**, offers a daring counter-narrative. It views SHA-256 as a structured, deterministic universe governed by "harmonic" laws. It redefines the halting signal as an internal "resonance" ("FOLD: TRUE") and proposes that preimages can be "created" through geodesic navigation.

While currently speculative and facing rigorous theoretical hurdles (specifically the "Continuum Fallacy"), the Logic paradigm represents the ultimate ambition of cryptanalysis: to replace the ignorance of the search with the certainty of the derivation. If the "halting signal" can indeed be internalized—if the system can "feel" when it is close to a solution—then the walls of the event horizon are not as impenetrable as we believe. The transition from "finding" to "creating" would mark the end of the current cryptographic age and the beginning of a new, resonant computational era.

Works cited

1. Deep Research Paper Drafting Protocol,
<https://drive.google.com/open?id=1yzq8rCNfvhoXIJl8NQQkczUgXPNmjoWNjRZKozJqero>
2. Programmatic SAT for SHA-256 Collision Attack, accessed January 8, 2026,
<https://uwindsor.scholaris.ca/bitstreams/64fe0597-3fc6-415f-99d8-d3e277f93ea7/download>
3. SHA Encryption Explained: SHA-1 vs. SHA-2 vs. SHA-3 | Sectigo® Official, accessed January 8, 2026, <https://www.sectigo.com/blog/what-is-sha-encryption>
4. Difficulty of collision vs preimage vs second-preimage attacks - Cryptography Stack Exchange, accessed January 8, 2026,
<https://crypto.stackexchange.com/questions/50223/difficulty-of-collision-vs-preimage-vs-second-preimage-attacks>

5. Are there two known inputs that give the same SHA256 output? : r/computerscience - Reddit, accessed January 8, 2026, https://www.reddit.com/r/computerscience/comments/12egpsu/are_there_two_known_inputs_that_give_the_same/
6. accessed January 8, 2026, <https://ftp.math.utah.edu/pub/tex/bib/imwut.bib>
7. A Deep Dive into SHA-256: Working Principles and Applications | by Madan | Medium, accessed January 8, 2026, https://medium.com/@madan_nv/a-deep-dive-into-sha-256-working-principles-and-applications-a38cccc390d4
8. Preimage attack - Wikipedia, accessed January 8, 2026, https://en.wikipedia.org/wiki/Preimage_attack
9. The Nexus Recursive Harmonic Framework: Formalizing Reality as Recursive Computation, accessed January 8, 2026, <https://zenodo.org/records/17983567>
10. The Nexus 4 Framework - Hybrid Optical Modulator Concept .docx, <https://drive.google.com/open?id=1Bm3HVs2uWfFWH5blilma8iSId2QthfMH>
11. Recursive Harmonic Architecture Emerges , <https://drive.google.com/open?id=1-gDgnPeqfx1vgF-vEl43ooOAhQuEAAONsaXKuBSvMAw>
12. I don't think double sha256 makes any difference with regards to collisions. If ... - Hacker News, accessed January 8, 2026, <https://news.ycombinator.com/item?id=39836877>
13. New Second-Preimage Attacks on Hash Functions - National Institute of Standards and Technology, accessed January 8, 2026, https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=918851
14. ELI5: Why is it that Sha256 cannot be reversible even though it always outputs the same result for the same input given : r/explainlikeimfive - Reddit, accessed January 8, 2026, https://www.reddit.com/r/explainlikeimfive/comments/15asggt/eli5_why_is_it_that_sha256_cannot_be_reversible/
15. Computational Complexity Theory - Stanford Encyclopedia of Philosophy, accessed January 8, 2026, <https://plato.stanford.edu/archives/fall2016/entries/computational-complexity/>
16. Computational requirements for breaking SHA-256? - Cryptography Stack Exchange, accessed January 8, 2026, <https://crypto.stackexchange.com/questions/52571/computational-requirements-for-breaking-sha-256>

17. Is SHA-256 Secure Against Quantum Attacks? | Post-Quantum Security Center: From VPN Vulnerabilities to Quantum Safe Victory - Gopher Security, accessed January 8, 2026, <https://www.gopher.security/post-quantum/is-sha-256-secure-against-quantum-attacks>
18. Quantum Computers Threat to Bitcoin | by Przemek Chojecki - Medium, accessed January 8, 2026, <https://pchojecki.medium.com/quantum-computers-threat-to-bitcoin-e1b57boda2aa>
19. SHA-256 Collision Attack with Programmatic SAT - arXiv, accessed January 8, 2026, <https://arxiv.org/html/2406.20072v1>
20. SHA-256: (Probabilistic?) partial preimage possible? - Cryptography Stack Exchange, accessed January 8, 2026, <https://crypto.stackexchange.com/questions/8786/sha-256-probabilistic-partial-preimage-possible>
21. SHA-256 Collision Attack with Programmatic SAT - CEUR-WS.org, accessed January 8, 2026, <https://ceur-ws.org/Vol-3717/paper5.pdf>
22. The Nexus Recursive Harmonic Intelligence Framework - Deriving a Universal Harmonic Phase Constant Across Scales - Zenodo, accessed January 8, 2026, <https://zenodo.org/records/18162886>